We create chemistry

Safety data sheet

1. Identification

Product identifier

Basoflux® RD 5119 T

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Performance Chemicals for Oilfield Applications

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Global Oilfield Chemicals

Telephone: +49 621 60-0
E-mail address: RegXcellenceOilfieldChemicals@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Flam. Liq. 4
STOT SE 3 (May cause drowsiness and dizziness.)
Aquatic Acute 3
Aquatic Chronic 3

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

H227	Combustible liquid.
H336	May cause drowsiness or dizziness.
H402	Harmful to aquatic life.
H412	Harmful to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P280	Wear protective gloves and eye protection or face protection.
P273	Avoid release to the environment.
P261	Avoid breathing mist or vapour or spray.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P370 + P378	In case of fire: Use foam or dry powder for extinction.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):
Repeated exposure may cause skin dryness or cracking.
Product contains the following components and may cause carcinogenicity: Naphthalene

Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Preparation based on: polyacrylate, solvent(s)

Hazardous ingredients (GHS)
According to UN GHS criteria

Distillates (petroleum), hydrotreated light

Content (W/W): >= 25 % - < 50 %	Asp. Tox. 1
CAS Number: 64742-47-8	Flam. Liq. 4
EC-Number: 265-149-8	H227, H304
INDEX-Number: 649-422-00-2	EUH066

solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified

Content (W/W): >= 20 % - < 25 %	Asp. Tox. 1
CAS Number: 64742-94-5	Flam. Liq. 4
EC-Number: 918-811-1	STOT SE 3 (drowsiness and dizziness)
	Aquatic Acute 2
	Aquatic Chronic 2
	H401, H411, H227, H304, H336
	EUH066

1,2,4-Trimethylbenzene

Content (W/W): >= 1 % - < 3 % CAS Number: 95-63-6 EC-Number: 202-436-9 REACH registration number: 01-211-9472135-42 INDEX-Number: 601-043-00-3	Asp. Tox. 1 Flam. Liq. 3 Acute Tox. 4 (Inhalation - vapour) Skin Irrit. 2 Aquatic Chronic 2 Eye Irrit. 2A STOT SE 3 (irr. to respiratory syst.) Aquatic Acute 2 H226, H319, H315, H332, H304, H335, H401, H411

naphthalene (as a component of solvent naphtha heavy)

Content (W/W): > 0 % - < 0,3 % CAS Number: 91-20-3 EC-Number: 202-049-5 INDEX-Number: 601-052-00-2	Flam. Sol. 2 Acute Tox. 4 (oral) Carc. 2 Aquatic Acute 1 Aquatic Chronic 1 M-factor acute: 1 M-factor chronic: 1 H228, H302, H351, H400, H410

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
dry powder, foam

Special hazards arising from the substance or mixture

harmful vapours, carbon oxides, hydrocarbons
Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Breathing protection required. Avoid inhalation. Avoid contact with skin. Ensure adequate ventilation. Sources of ignition should be kept well clear.

Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Precautions for safe handling

Ensure thorough ventilation of stores and work areas. Avoid skin contact. Avoid inhalation of the product. Wear suitable personal protective equipment.

No eating, drinking, smoking or tobacco use at the place of work. Remove contaminated clothing and protective equipment before entering eating areas. Wash hands before breaks and at end of work.

Protection against fire and explosion:
Sources of ignition should be kept well clear. Take precautionary measures against static discharges. If delivered in plastic packing, highest permissable emptying temperature is 5 Kelvin below the flash point.

Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds. Segregate from nitrites.
Further information on storage conditions: Keep container tightly closed and in a cool place.

Storage stability:
Storage temperature: <= 50 °C

The packed product is not damaged by low temperatures or by frost.
Protect from temperatures above: 50 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

91-20-3: Naphthalene
95-63-6: 1,2,4-Trimethylbenzene
64742-47-8: Distillates (petroleum), hydrotreated light
64742-94-5: Solvent naphtha (petroleum), heavy arom.
64742-47-8: Distillates (petroleum), hydrotreated light

Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Gas filter for gases/vapours of organic compounds (boiling point >65 °C, e. g. EN 14387 Type A)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
fluoroelastomer (FKM) - 0.7 mm coating thickness
Polyethylene-Laminate (PE laminate) - ca. 0.1 mm coating thickness
Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN ISO 374-1)
nitrile rubber (NBR) - 0.4 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.
Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Wearing of closed work clothing is required additionally to the stated personal protection equipment.
Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid to waxy	
Colour:	colourless to yellowish	
Odour:	mineral oil-like	
Odour threshold:	not determined	
Melting point:	17 - 23 °C	
Boiling point:	> 190 °C	
Flammability:	Combustible liquid.	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Flash point:	62,5 °C The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.	(ISO 2719)
Auto-ignition temperature:	> 230 °C	(DIN 51794)
Thermal decomposition:	375 °C, > 200 J/g (DSC (DIN 51007)) self-accelerating reaction	
pH value:	not applicable, insoluble	
Viscosity, kinematic:	approx. 1.060 mm2/s (23 °C)	(calculated (from dynamic viscosity))
Viscosity, dynamic:	932 mPa.s (23 °C)	
Solubility in water:	dispersible when hot	
Solubility (qualitative) solvent(s):	aromatic hydrocarbons soluble	
Partitioning coefficient n-octanol/water (log Kow):	not applicable for mixtures	
Vapour pressure:	approx. 5,96 hPa (20 °C) static	

	approx. 17,82 hPa (50 °C) static	
	approx. 20,98 hPa (55 °C) static	
Density:	0,89 g/cm3 (15 °C)	(ISO 2811-3)
	0,879 g/cm3 (20 °C)	(ISO 2811-3)
	0,8581 g/cm3 (50 °C)	(ISO 2811-3)
Relative vapour density (air):	> 1 (20 °C) Heavier than air.	(estimated)

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

Corrosion to metals

No corrosive effect on metal.

Other safety characteristics

Hygroscopy: Non-hygroscopic

Evaporation rate:

not determined

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Peroxides: 0 %
The product does not contain peroxides.

Possibility of hazardous reactions

No hazardous reactions when stored and handled according to instructions.

Conditions to avoid

Avoid all sources of ignition: heat, sparks, open flame. Avoid electro-static discharge.

Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Experimental/calculated data:
ATE (oral): > 5.000 mg/kg

ATE (dermal): > 5.000 mg/kg

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified

Experimental/calculated data:
LD50 rat (oral): 6.318 mg/kg (OECD Guideline 401)

Information on: Distillates (petroleum), hydrotreated light

Experimental/calculated data:
LD50 rat (oral): > 5.000 mg/kg (OECD Guideline 420)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Experimental/calculated data:
LC50 rat (by inhalation): > 4,688 mg/l (OECD Guideline 403)

Information on: Distillates (petroleum), hydrotreated light
Experimental/calculated data:
LC50 rat (by inhalation): > 5,28 mg/l 4 h (OECD Guideline 403)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The vapour was tested.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Experimental/calculated data:

LD50 rabbit (dermal): > 2.000 mg/kg (OECD Guideline 402)

Information on: Distillates (petroleum), hydrotreated light
Experimental/calculated data:
LD50 rabbit (dermal): > 2.000 mg/kg (OECD Guideline 402)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Irritation

Assessment of irritating effects:
May cause slight irritation to the skin. May cause slight irritation to the eyes. May cause slight irritation to the respiratory tract.

Respiratory/Skin sensitization

Assessment of sensitization:
Based on available data, the classification criteria are not met.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of sensitization:
No sensitizing effect.

Information on: Distillates (petroleum), hydrotreated light
Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on the ingredients, there is no suspicion of a mutagenic effect.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

Information on: Distillates (petroleum), hydrotreated light
Assessment of mutagenicity:
In the majority of tests performed (bacteria/microorganisms/cell cultures) a mutagenic effect was not found. A mutagenic effect was also not observed in in-vivo assays. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Information on: Distillates (petroleum), hydrotreated light
Assessment of carcinogenicity:
Long-term exposure to highly irritating concentrations resulted in skin tumors in animals. A carcinogenic effect in humans can be excluded after brief skin contact. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

Information on: Distillates (petroleum), hydrotreated light
Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:
Based on the ingredients, there is no suspicion of a teratogenic effect.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Assessment of teratogenicity:
Based on available data, the classification criteria are not met.

Information on: Distillates (petroleum), hydrotreated light
Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Possible narcotic effects (drowsiness or dizziness).

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components. Has a degreasing effect on skin.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. Harmful to aquatic life with long lasting effects.

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Toxicity to fish:
LL50 (96 h) 2 mg/l, Oncorhynchus mykiss

LL50 (96 h) < 5 mg/l, Oncorhynchus mykiss

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Aquatic invertebrates:
EL50 (48 h) 3 mg/l, Daphnia magna

EL50 (48 h) < 10 mg/l, Daphnia magna

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Chronic toxicity to fish:
NOEL (28 d) 0,441 mg/l, Oncorhynchus mykiss

Information on: solvent naphtha (petroleum), heavy arom.; Kerosine - unspecified
Chronic toxicity to aquatic invertebrates:
NOEL (21 d) 0,771 mg/l, Daphnia magna

Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Bioaccumulative potential

Assessment bioaccumulation potential:
Accumulation in organisms is expected.

Information on: Distillates (petroleum), hydrotreated light
Assessment bioaccumulation potential:
Accumulation in organisms is expected.

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: No data available.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components. Do not discharge product into the environment without control.

13. Disposal Considerations

Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable

Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable

UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Flam. Liq.	Flammable liquids
STOT SE	Specific target organ toxicity — single exposure
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Asp. Tox.	Aspiration hazard
Acute Tox.	Acute toxicity
Skin Irrit.	Skin irritation
Eye Irrit.	Eye irritation
Flam. Sol.	Flammable solids
Carc.	Carcinogenicity
H227	Combustible liquid.
H304	May be fatal if swallowed and enters airways.
H401	Toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.
H336	May cause drowsiness or dizziness.
H226	Flammable liquid and vapour.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H335	May cause respiratory irritation.
H228	Flammable solid.
H302	Harmful if swallowed.
H351	Suspected of causing cancer.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
EUH066	Repeated exposure may cause skin dryness or cracking.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the

responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.